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SOFTWARE PROJECT MANAGEMENT PLAN

Hybrid Deep Learning Based Breast Cancer Classification Model Using Mammographic and Sonographic Images

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Submission Date	March 2026

Prepared in accordance with IEEE Std 1058-1998 (Software Project Management Plans), PMBOK® Guide (7th Edition) principles, and ISO/IEC 12207:2017 software lifecycle processes.

DECLARATION

We hereby declare that this Software Project Management Plan has been prepared independently by the undersigned as part of our Final Year Project at Bahria University, Karachi Campus. All information herein is based on original work, except for citations and referenced materials which have been duly acknowledged. This document has not been previously submitted for any academic award at this or any other institution.

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1 INTRODUCTION

1.1 Purpose of this SPMP

This Software Project Management Plan (SPMP) is prepared in accordance with IEEE Std 1058-1998, PMBOK® Guide (7th Edition) principles, and ISO/IEC 12207:2017 software lifecycle processes. It provides a comprehensive framework governing the planning, execution, monitoring, and closure of the Final Year Project titled *Hybrid Deep Learning Based Breast Cancer Classification Model Using Mammographic and Sonographic Images*. The document formalises project scope, schedule, resource allocation, risk strategy, quality assurance practices, and communication protocols for all stakeholders.

1.2 Project Overview

Breast cancer remains a leading cause of cancer-related mortality among women worldwide, necessitating early and precise detection to improve survival outcomes. Contemporary deep learning systems predominantly adopt *uni-modal* approaches — processing either mammographic or sonographic images in isolation — thereby forfeiting complementary diagnostic information available across modalities.

This project proposes a **Hybrid Multimodal Deep Learning Framework** that simultaneously processes paired mammography and ultrasound images through parallel Convolutional Neural Network (CNN) branches. Features from each branch are integrated via a cross-modal attention-based fusion layer, and the resulting unified representation is used to classify breast lesions as benign or malignant. The system is benchmarked against state-of-the-art uni-modal baselines using standard medical imaging metrics.

1.3 Project Deliverables

- A functional hybrid deep learning framework integrating mammographic and sonographic data.
- Fully preprocessed and augmented dataset pipelines for the CBIS-DDSM and BUSI datasets.
- Trained uni-modal CNN baseline models for mammography and ultrasound independently.
- Attention-based multimodal fusion architecture inspired by TransBreastNet.
- Comparative performance evaluation report against state-of-the-art models.
- Final thesis documentation and academic presentation for viva examination.
- Version-controlled source code repository with full reproducibility documentation.

1.4 Assumptions and Constraints

Category	Description
Assumption	CBIS-DDSM mammography and BUSI ultrasound datasets are publicly available and accessible via TCIA and Kaggle respectively.
Assumption	GPU-accelerated hardware (NVIDIA RTX series or Google Colab Pro / Kaggle TPU) is available throughout the project.
Assumption	Both team members maintain concurrent and active progress throughout all applicable project phases.
Constraint	The project must be completed within the academic year: January 2026 – December 2026.
Constraint	The system is a research prototype only; clinical deployment is explicitly out of scope.
Constraint	All development uses open-source Python libraries (TensorFlow / PyTorch, OpenCV, etc.).
Constraint	Binary classification (Benign vs. Malignant) is the primary objective; multi-class is a stretch goal only.

2 PROJECT ORGANISATION

2.1 Project Team Structure

The project is executed by a two-member student team supervised by an academic faculty member. The structure follows a *flat project team model* consistent with small-scale academic research. All technical decisions are made collaboratively with formal milestone approval from the supervising faculty.

Role	Name	ID / Designation	Primary Responsibilities
Project Supervisor	Miss Fatima Bashir	Faculty, CS Dept., BU Karachi	Academic oversight, technical guidance, milestone sign-off, thesis approval
Project Lead / ML Engineer	S M Aun Abidi	02-134231-094	Architecture design, model training, fusion implementation, data acquisition, integration, primary documentation
Research & ML Engineer	Tarique Ahmed	02-134231-011	Preprocessing pipeline support, literature review, evaluation, comparative analysis, thesis co-authorship

Project Supervisor — Miss Fatima Bashir

Provides academic and technical direction; formally reviews and approves all major milestone deliverables; conducts bi-weekly progress meetings; approves the final thesis for examination.

Project Lead / ML Engineer — S M Aun Abidi

Leads all phases of model development: data acquisition (CBIS-DDSM and BUSI datasets), CNN architecture design, attention-based fusion implementation, and ablation studies. Coordinates overall project activities and authors primary documentation sections.

Research & ML Engineer — Tarique Ahmed

Supports the preprocessing pipeline, conducts and maintains the literature review, performs comparative evaluation against published benchmarks, contributes to thesis writing, and prepares the final viva defence presentation.

3 MANAGERIAL PROCESS PLANS

3.1 Work Breakdown Structure (WBS)

The project is decomposed into six phases with discrete work packages. The WBS follows PMBOK hierarchical decomposition. Each task carries an explicit owner, start/end dates, and live status indicator drawn from the Instagantt tracker.

ID	Phase / Work Package	Owner	Start	End	Status
1.0	**PHASE 1 — PLANNING & LITERATURE REVIEW**	**Both**	**Jan 2026**	**Feb 2026**	**Finished**
1.1	Literature Review Initiation	Aun	14 Jan	01 Feb	Finished
1.2	Project Proposal Preparation	Both	01 Feb	20 Feb	Finished
1.3	Proposal Defense & Approval	Both	20 Feb	28 Feb	Finished
2.0	**PHASE 2 — DATA ACQUISITION & PREPROCESSING**	**Aun**	**01 Mar**	**01 May**	**Started**
2.1	DDSM Dataset Licensing & Download	Aun	01 Mar	20 Mar	Finished
2.2	BUSI Dataset Research & Acquisition	Aun	20 Mar	15 Apr	Finished
2.3	Image Normalisation (Intensity & Spatial)	Aun	15 Apr	25 Apr	Started
2.4	Noise Reduction (Gaussian & Median Filtering)	Aun	25 Apr	01 May	Not Started
3.0	**PHASE 3 — UNI-MODAL MODEL TRAINING**	**Both**	**01 May**	**30 Jun**	**Not Started**
3.1	Mammography CNN — Architecture Design	Both	01 May	15 May	Not Started
3.2	Mammography CNN — Training Implementation	Both	15 May	30 May	Not Started
3.3	Ultrasound CNN — Architecture Design	Both	30 May	15 Jun	Not Started
3.4	Ultrasound CNN — Training Implementation	Both	15 Jun	30 Jun	Not Started
4.0	**PHASE 4 — HYBRID ARCHITECTURE DEVELOPMENT**	**Both**	**01 Jul**	**31 Aug**	**Not Started**
4.1	Feature Extraction & Attention Mechanism Design	Both	01 Jul	30 Jul	Not Started
4.2	Integration Testing & Feature Alignment	Both	30 Jul	15 Aug	Not Started
4.3	Hybrid Model Performance Validation	Both	15 Aug	31 Aug	Not Started
5.0	**PHASE 5 — OPTIMISATION & TESTING**	**Both**	**01 Sep**	**31 Oct**	**Not Started**
5.1	Grid Search Hyperparameter Tuning	Both	01 Sep	15 Sep	Not Started
5.2	Bayesian Optimisation	Both	15 Sep	30 Sep	Not Started
5.3	Cross-modal Refinement & Ablation Studies	Both	30 Sep	15 Oct	Not Started
5.4	Final Model Validation	Aun	15 Oct	31 Oct	Not Started
6.0	**PHASE 6 — DOCUMENTATION & DEFENCE**	**Both**	**01 Nov**	**31 Dec**	**Not Started**

ID	Phase / Work Package	Owner	Start	End	Status
6.1	Thesis Writing	Aun	01 Nov	30 Nov	Not Started
6.2	Final Presentation Preparation	Both	30 Nov	15 Dec	Not Started
6.3	Final Defence (Viva Examination)	Both	15 Dec	31 Dec	Not Started

*Status Legend: **Finished** · **Started** · **Not Started***

3.2 Project Schedule and Timeline

The project spans January 2026 to December 2026 across six sequential phases. The Gantt chart below is the actual export from the Instagantt project management platform (generated 11 March 2026), showing all tasks, assignments, and live progress indicators against the planned timeline.

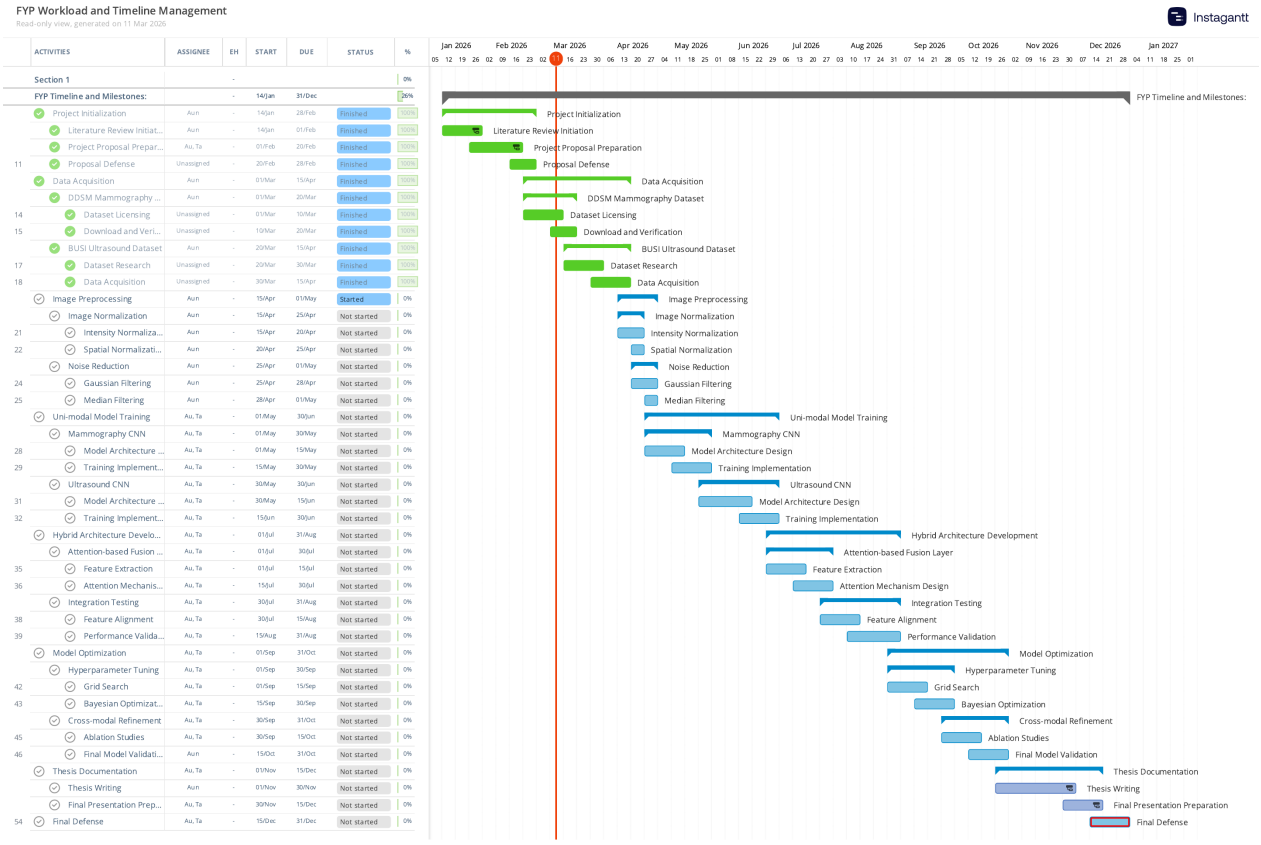

Figure 1: FYP Workload and Timeline Management — Gantt Chart (Instagantt, 11 Mar 2026)

Key Project Milestones

Milestone	Description	Target Date	Status
M1 – Proposal Approved	Project proposal defended and formally approved	Feb 28, 2026	Achieved
M2 – Datasets Ready	CBIS-DDSM and BUSI datasets acquired and fully preprocessed	Apr 30, 2026	In Progress
M3 – Baselines Trained	Uni-modal CNNs trained and benchmarked vs. published results	Jun 30, 2026	Not Started
M4 – Hybrid Prototype	Attention-based fusion architecture implemented and tested	Aug 31, 2026	Not Started
M5 – Model Optimised	Hyperparameter tuning and ablation studies completed	Oct 31, 2026	Not Started
M6 – Thesis Submitted	Complete thesis submitted to examination committee	Nov 30, 2026	Not Started
M7 – Project Defence	Final viva examination and presentation delivered	Dec 31, 2026	Not Started

3.3 Resource Management Plan

The following resources are required across the project lifecycle. Compute resources are pre-allocated at the start of each training phase to avoid scheduling conflicts.

Category	Resource	Phase(s)	Availability
Human Resources	S M Aun Abidi — Project Lead	All phases	Full-time (FYP)
Human Resources	Tarique Ahmed — ML Engineer	All phases	Full-time (FYP)
Human Resources	Miss Fatima Bashir — Supervisor	All phases	Bi-weekly reviews
Compute (Local)	GPU Workstation (NVIDIA RTX series)	Phases 3–5	As available
Compute (Cloud)	Google Colab Pro / Kaggle TPU	Phases 3–5	On demand
Dataset	CBIS-DDSM Mammography (TCIA)	Phase 2 onward	Public (CC BY 3.0)
Dataset	BUSI Ultrasound (Kaggle)	Phase 2 onward	Public (Research)
Software	Python 3.10+, TensorFlow / PyTorch 2.x	All phases	Open source
Software	OpenCV, scikit-learn, NumPy, Pandas, Optuna	All phases	Open source
Version Control	Git + GitHub (private repository)	All phases	Free tier
Documentation	LaTeX / Overleaf	Phase 6	University licence

3.4 Risk Management Plan

Risk management follows the PMBOK cycle: **Identify** → **Analyse** → **Plan Response** → **Monitor**. Probability and Impact are rated Low (1), Medium (2), or High (3). The Risk Priority Number (RPN) is their product. RPN ≥ 6 demands active mitigation.

ID	Risk Description	Prob	Imp	RPN	Mitigation Strategy
R-01	GPU hardware unavailability or failure	Med	High	6	Pre-book Google Colab Pro / Kaggle as cloud fallback; schedule compute slots in advance.
R-02	DDSM dataset licensing delay or access denial	Low	High	3	Prepare alternatives (INbreast, VinDr-Mammo); submit access request early in Phase 2.
R-03	Model convergence failure or severe overfitting	Med	High	6	Apply dropout and L2 regularisation; use data augmentation; employ early stopping callbacks.
R-04	Team member unavailability (illness / commitments)	Low	Med	2	Cross-train both members on all modules; maintain thorough inline code documentation.
R-05	Scope creep into multi-class classification	Med	Med	4	Strictly prioritise binary classification; treat multi-class as a post-defence extension only.
R-06	No improvement over uni-modal baseline performance	Med	High	6	Explore multiple fusion strategies (early, late, hybrid); document negative results as valid research.
R-07	Severe class imbalance (benign vs. malignant)	High	High	9	Apply SMOTE oversampling, class-balanced batch sampling, and weighted cross-entropy loss.
R-08	Timeline slippage due to iterative experimentation	High	Med	6	Build two-week buffers into each phase; track weekly on Gantt; escalate delays to supervisor.
R-09	Loss of code or experimental results	Low	High	3	Daily Git commits to remote repository; automatic checkpoint backup to Google Drive.

*RPN = Probability × Impact (High=3, Med=2, Low=1) · **RPN ≥ 6 = High Risk** · **RPN 4–5 = Medium** · **RPN ≤ 3 = Low***

4 TECHNICAL PROCESS PLANS

4.1 Development Methodology

The project adopts an **Iterative and Incremental** methodology consistent with ISO/IEC 12207 lifecycle processes. It blends the *Spiral Model* (risk-driven iteration on model architecture) with *Agile Scrum* sprint cadence (weekly planning and review). This is well-suited to machine learning research where experimental outcomes drive subsequent design decisions.

Phase 1 – Planning & Literature Review

Systematic review of multimodal deep learning in medical imaging. Identification of benchmark datasets, baseline architectures, and standard evaluation metrics. Formal proposal preparation and supervisor-approved scope definition.

Phase 2 – Data Acquisition & Preprocessing (Owner: Aun)

Procurement of CBIS-DDSM and BUSI datasets. Construction of the preprocessing pipeline: intensity and spatial normalisation, Gaussian/Median noise filtering, and augmentation (flipping, rotation, brightness jitter). Tarique provides pipeline support.

Phase 3 – Uni-modal Model Training

Independent CNN training on CBIS-DDSM (mammography) and BUSI (ultrasound) using ImageNet-pretrained transfer learning backbones. Benchmarking against published state-of-the-art results to establish quantitative baselines.

Phase 4 – Hybrid Architecture Development

Core innovation phase: parallel CNN branches connected via a cross-modal attention-based fusion layer inspired by TransBreastNet. Feature alignment and integration testing performed iteratively until stable convergence is achieved.

Phase 5 – Optimisation & Testing

Hyperparameter search via Grid Search and Bayesian Optimisation (Optuna). Ablation studies quantifying each modality's independent contribution. Final evaluation on held-out test sets.

Phase 6 – Documentation & Defence

Comprehensive thesis writing per Bahria University formatting guidelines. Preparation of the final defence presentation and viva materials.

4.2 System Architecture Overview

The proposed system follows a modular five-component pipeline. Each component is independently unit-testable, ensuring traceability from requirements through to implementation and evaluation results.

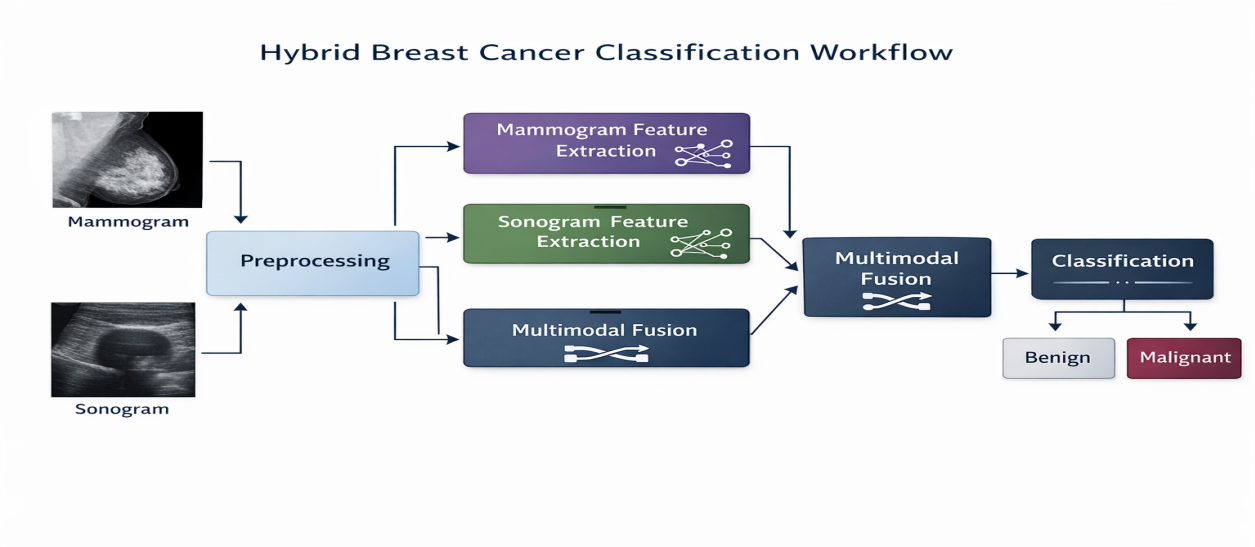

Figure 2: Hybrid Breast Cancer Classification Workflow — Preprocessing → Dual CNN Extraction → Attention Fusion → Classification

Data Preprocessing Module

Resizes images to a uniform spatial dimension, applies intensity normalisation, Gaussian/Median noise filtering, and performs augmentation. Outputs standardised input tensors for both CNN branches.

Mammography Feature Extraction Module

A CNN backbone (ResNet50 / EfficientNet-B4) pretrained on ImageNet extracts high-level spatial and texture features from mammograms via transfer learning.

Sonography Feature Extraction Module

A dedicated CNN backbone processes ultrasound images to extract morphological and echogenicity-based features, handling noise variability inherent in sonography.

Attention-Based Fusion Module

A cross-modal attention mechanism selectively weights feature maps from both branches, suppresses irrelevant activations, and produces a unified multimodal representation.

Classification Module

Fully connected layers map the fused feature vector to class probabilities (Benign / Malignant) via Softmax activation.

4.3 Technology Stack

Category	Technology / Tool	Purpose
Language	Python 3.10+	Primary development language
Deep Learning	TensorFlow 2.x / PyTorch 2.x	Model architecture and training
Image Processing	OpenCV 4.x	Preprocessing and augmentation pipeline
Data Science	NumPy, Pandas, Matplotlib, Seaborn	Data handling and result visualisation
ML Utilities	scikit-learn	Cross-validation, metrics, classical ML
Hyperparameter Opt.	Optuna / scikit-optimize	Grid Search and Bayesian Optimisation
Compute (Local)	NVIDIA RTX GPU Workstation	Accelerated model training
Compute (Cloud)	Google Colab Pro / Kaggle TPU	Fallback and parallel experimentation
Version Control	Git + GitHub (private repository)	Code versioning and team collaboration
Datasets	CBIS-DDSM, BUSI	Primary training and evaluation data
Documentation	LaTeX / Overleaf	Thesis and formal report writing
Project Tracking	Instagantt	Task scheduling and progress monitoring

4.4 Data Management Plan

Both datasets are publicly available and contain no personally identifiable patient information. Data is stored locally with automatic backup to Google Drive.

Dataset	Modality	Source	Size	Licence
CBIS-DDSM	Mammography	The Cancer Imaging Archive (TCIA)	~2,620 cases	CC BY 3.0
BUSI	Ultrasound	Kaggle / Al-Dhabyani et al. 2020	~780 images	Public Research

All preprocessing scripts are version-controlled in the GitHub repository. Experimental artefacts (training logs, model checkpoints, confusion matrices) follow the naming convention: *YYYYMMDD_experiment_description/*.

5 QUALITY ASSURANCE PLAN

5.1 Quality Objectives

- Achieve classification accuracy ≥ 90% on held-out test set for the hybrid model.
- Demonstrate statistically significant improvement over uni-modal baselines.
- Full reproducibility: all experiments replicable from the GitHub repository.
- All source code documented with inline comments and PEP 257 docstrings.
- All deliverables reviewed and formally approved by supervisor before submission.
- Each pipeline module independently unit-testable with no single point of failure.

5.2 Verification and Validation (V&V;)

The project follows IEEE Std 1012-2016 V&V; processes. **Verification** confirms each component is built correctly per specification; **Validation** confirms the integrated system meets stated research objectives.

V&V; Activity	Phase	Method	Acceptance Criterion
Dataset Integrity Check	Phase 2	Statistical analysis of class distribution, image dimensions, and corrupt file detection	No corrupt files; class imbalance documented and addressed
Preprocessing Validation	Phase 2	Visual inspection and pixel statistics before and after normalisation	Pixel range normalised to [0, 1]; measurable SNR improvement
Uni-modal Baseline Validation	Phase 3	5-fold cross-validation; results compared against published benchmark values	Performance within ±2% of reported literature values
Hybrid Integration Test	Phase 4	Unit tests per module; end-to-end inference test on a held-out sample batch	Zero runtime errors; correct output shape and class labels
Final Performance Validation	Phase 5	Independent test-set evaluation; ablation study across fusion strategies	AUC ≥ 0.90; Sensitivity ≥ 88%; Specificity ≥ 85%
Documentation Review	Phase 6	Supervisor review of thesis draft; peer proofreading by both team members	All comments resolved; formatting compliant with university template

5.3 Performance Metrics

All metrics below are computed for both uni-modal baselines and the final hybrid model on the same held-out test set to ensure fair comparison.

Metric	Formal Definition	Target
Accuracy	Correctly classified samples ÷ Total samples	≥ 90%
Sensitivity (Recall)	TP ÷ (TP + FN) — clinically critical: must not miss malignancies	≥ 88%
Specificity	TN ÷ (TN + FP)	≥ 85%
AUC-ROC	Area under the Receiver Operating Characteristic Curve	≥ 0.90
F1-Score	2 × (Precision × Recall) ÷ (Precision + Recall)	≥ 0.88
Precision	TP ÷ (TP + FP)	Reported
Confusion Matrix	Full breakdown: True/False Positives and True/False Negatives	Reported

6 CONFIGURATION MANAGEMENT PLAN

6.1 Version Control Strategy

All source code, configuration files, and experiment scripts are managed via **Git** with a private **GitHub** repository. The branching strategy follows a simplified Git Flow to maintain stability while enabling parallel work.

Branch	Purpose	Merge Policy
main	Stable checkpoints approved by supervisor	Supervisor approval required
develop	Active development integration branch	Peer code review by both members via Pull Request
feature/*	Individual module development (e.g. feature/fusion-layer)	Merge into develop via Pull Request
experiment/*	Isolated model training and hyperparameter experiments	Results documented before merging
hotfix/*	Critical bug fixes on stable production code	Direct merge to both main and develop

6.2 Document Control

All project documents are versioned as **DocumentName_vX.Y_YYYYMMDD** and stored in a shared Google Drive folder with supervisor read access. Baselines are established at each formal milestone approval.

Document	Ver.	Date	Status
FYP Proposal Report	v1.0	Feb 2026	Baselined
Software Project Management Plan (SPMP)	v1.0	Mar 2026	Current
Preprocessing Technical Specification	v0.1	Apr 2026	Pending
Model Architecture Design Document	v0.1	Jul 2026	Pending
Evaluation and Results Report	v0.1	Oct 2026	Pending
Final Thesis	v1.0	Nov 2026	Pending
Final Defence Presentation	v1.0	Dec 2026	Pending

7 COMMUNICATION AND REPORTING PLAN

Structured communication is essential for maintaining project momentum and alignment between team members and the supervising faculty. The plan below is adopted for the full project duration.

Communication Type	Frequency	Participants	Channel	Output / Artefact
Team Stand-up	Weekly	Both members	In-person / WhatsApp	Progress log; sprint task list for next week
Supervisor Meeting	Bi-weekly	Both members + Miss Fatima Bashir	In-person — Bahria University	Meeting minutes; formal milestone sign-off
Code Review	Per feature branch	Both members	GitHub Pull Requests	Reviewed, approved, and merged code
Milestone Report	Per milestone	Both members	Written report to supervisor	Milestone completion record
Experiment Log Update	Per experiment	Responsible member	Shared Google Drive	Timestamped experiment log with results
Emergency Escalation	As required	Both members + Supervisor	Email / Phone call	Agreed action plan within 48 hours

Progress Tracking Tools

The **Instagantt Gantt chart** is the primary tracking tool, updated weekly to reflect actual vs. planned progress. **GitHub Insights** (commit history and issue tracker) provides secondary monitoring. A **shared Google Drive** folder stores all meeting minutes, milestone reports, and versioned documents.

8 PROJECT CLOSURE AND EVALUATION

8.1 Project Closure Criteria

The project is formally closed when all of the following conditions are satisfied:

- All six phases completed and signed off by the supervisor.
- Final model trained, validated, and documented with fully reproducible results.
- Complete thesis submitted and accepted by the examination committee.
- Source code committed to GitHub with README and reproducibility guide.
- Final viva examination successfully passed and graded.

8.2 Lessons Learned Review

Upon project completion a structured lessons-learned session will capture: (a) what worked well in development and management; (b) challenges encountered and how they were resolved; (c) recommendations for future multimodal deep learning FYP projects at Bahria University Karachi Campus. Outputs will be archived for departmental reference.

8.3 Research Impact and Future Directions

The proposed framework has potential for broader academic and clinical impact. Future research directions identified through the literature review include:

- Extension to multi-class classification: Normal / Benign / Malignant / DCIS.
- Incorporation of additional modalities such as breast MRI or contrast-enhanced CT.
- Clinical deployment study in partnership with a healthcare institution.
- Integration of Explainable AI (Grad-CAM, SHAP) to support radiologist trust and interpretability.
- Submission of results to a peer-reviewed venue (MICCAI, ISBI, or a Q1 journal).
- Extension of the attention-fusion strategy to 3D volumetric imaging data.

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End of Document · Software Project Management Plan v1.0 · March 2026
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